Pre-Development Framework

For custom web application V2

Instructions for Participants : Please answer the questions below in simple words. There are no "right or wrong" answers. Your answers will help us clearly understand your business needs and design the right application for you.

1. **.What is the main problem you're facing today?**

   Tip: Describe the biggest pain point in your daily work

Dhanashree Crop Solutions currently manages a field force of approximately 60 agronomists and crop advisors, each responsible for 600–2,000 farmers. All interactions — farm visits, phone calls, recommendations, crop data, and promotional communications — are tracked through manual, disconnected processes. The core problems are:

- No central repository: farmer records, visit logs, and crop-stage data are scattered across paper registers, personal Excel sheets, and WhatsApp chats belonging to individual staff members.
- No real-time management visibility: managers cannot monitor which farmers have been visited, which are overdue, or how field staff are performing — without physically calling each person.
- No automated reminders: there is no system to alert a field staff member when a farmer is overdue for a visit or when a crop is approaching its next growth stage.
- No targeted content delivery: there is no structured mechanism to push the right promotional material (product videos, advisories) to farmers at the correct crop-growth stage.
- No audit trail: there is no immutable record of the recommendations given or messages sent to farmers, creating compliance and quality-review gaps.
- No data for planning: management cannot aggregate data by crop type, village, territory, or growth stage to make informed inventory or campaign decisions.

In summary, the business is operating a 60-person field force and 50,000-farmer base entirely on informal, manually maintained information — making it impossible to scale, monitor, or optimise farmer engagement.

2. **What is the ONE most important thing this app should help you do?**

   Tip: If the app could only do one thing perfectly, what would it be?

The single most critical function of the FFMA is:

> *Give every field staff member a mobile-first, offline-capable digital workspace where they can manage their entire farmer portfolio — logging visits, tracking crop stages, and communicating recommendations — while giving managers real-time visibility into all field activity across every territory.*

If the app could do only one thing perfectly, it would be: ensure that no farmer is ever forgotten — by surfacing exactly who needs a visit today, what stage their crops are at, and what content to share with them — and logging every interaction permanently so management always knows what is happening in the field.

3. **How do you handle this work TODAY?**

   Tip: Describe your current process: Manual? Excel sheets? Paper registers? WhatsApp? Combination?

The current process is entirely manual and fragmented:

Farmer Records	Paper registers and personal Excel sheets maintained by each field staff member independently. No standardised format
Visit Logging	Field staff mentally track or note on paper which farmers they visited and when. There is no digital timestamping
Crop-Stage Tracking	No formal tracking. Staff rely on memory or handwritten notes to remember which crop stage each farmer's plot is at.
Recommendations	Shared verbally during farm visits or via personal WhatsApp messages. No structured record of product name, dose, or timing.

Promotional Content	Staff forward marketing materials from their personal WhatsApp to farmers individually. No tracking of who received what, or when.
Call Records	Not logged at all. There is no record of calls made, their duration, or outcomes.
Management Oversight	Managers must phone each field staff member individually to get updates. No dashboard or report exists.
Overdue Visit Alerts	No system. Farmers can go unvisited for weeks or months without any alert being generated.
Bulk Import / Data	Approximately 50,000 existing farmer records are held in Excel files, not accessible on-device by field staff.

The net result is that every piece of information lives in a person's head, a paper notebook, or a private phone — making the business completely dependent on individual memory and impossible to audit or scale.

4. Who will use this app?

List all types of users (e.g., owner, staff, customers, vendors, delivery persons)

The application has five distinct user types:

User Type	Approx. Count	Access Scope
Field Staff / Agronomist	~60	Own assigned farmers only
Territory Manager	~10–15	All staff and farmers within their territory

Regional / Zonal Manager	~3–5	All territories within their zone
System Administrator	1–2	Full system access — all data, config, and master data
Marketing / Content Team	2–5	Content library only — upload and tag promotional material

5. **What will each user type mainly do?**

Tip: For each user type listed above, describe their main task.
Example Format:
Owner → View reports and approve refunds
Staff → Create orders and update status
Customer → Place orders and track delivert

Field Staff / Agronomist

- Search and open farmer profiles from their assigned portfolio of 600–2,000 farmers.
- Add new farmers on the go (during field visits).
- Record the crop and variety grown on each plot, along with the sowing date and current growth stage.
- Log field visits with date, time, GPS location, notes, and photos.
- Log calls made to or received from farmers, including outcome notes.
- Create and send product recommendations (product name, dose, timing) via WhatsApp or SMS.
- Push crop-stage-relevant promotional content (videos, images, links) to individual farmers or bulk groups.
- Use the Smart Visit Planner to plan their daily visit schedule based on overdue status and GPS proximity.

Territory Manager

- Monitor all field staff activity in their territory — visits made, calls logged, recommendations sent.
- Review overdue visit lists and escalate to relevant staff.
- View farmer-level data across all staff in the territory (read-only).
- Access territory-level dashboards: farmers by village, crop, stage; staff leaderboard; promotion reach.
- Approve or review recommendations made by field staff for quality assurance.

Regional / Zonal Manager

- Access aggregated reports across multiple territories in the zone.
- Review staff performance across the zone via the Staff Leaderboard.
- Initiate or approve bulk campaign sends across the zone.
- Drill down from zone → territory → individual staff → individual farmer activity.

System Administrator

- Create and manage all user accounts and assign roles.
- Configure and restructure the territory hierarchy (Region → Zone → Territory → Staff → Farmer).
- Manage Crop Master data: add/edit crops, varieties, growth stages, and inter-stage durations.
- Execute bulk farmer imports from Excel (up to ~50,000 records).
- Configure WhatsApp Business API and SMS gateway credentials.
- Set visit frequency norms per farmer segment.
- Access and export audit logs.

Marketing / Content Team

- Tag each item to specific crops, growth stages, and languages.
- Attach pre-composed WhatsApp/SMS message templates to each content item.
- Activate or deactivate farmers and content items without deletion.

6. What can each user type do and NOT do?

Tip: Define permissions clearly
Example:
Admin: Can do everything including deleting data
Staff: Can create/edit orders but cannot delete or see financial reports
Customer: Can only view their own orders

Permissions are enforced at the API layer — not just the UI — so role boundaries cannot be bypassed regardless of the device or API call used.

Action / Capability	Field Staff	Territory Mgr	Zonal Mgr	Admin
View own farmers only	✅ Yes	✅ Full territory	✅ Full zone	✅ All
Add / edit farmer records	✅ Yes	✅ Yes	❌ No	✅ Yes
Delete farmer records	❌ No	❌ No	❌ No	✅ Yes
Log visits	✅ Yes	✅ Yes	❌ No	❌ No
Log calls	✅ Yes	✅ Yes	❌ No	❌ No
Create recommendations	✅ Yes	✅ Yes	❌ No	❌ No
Send WhatsApp / SMS to farmers	✅ Yes	✅ Yes	✅ Bulk send	✅ Yes

Delete message history	❌ No	❌ No	❌ No	❌ No
View territory-level reports	❌ No	✅ Yes	✅ Yes	✅ Yes
View financial / bulk export data	❌ No	✅ Yes	❌ No	✅ Yes
Upload / manage promotional content	❌ No	❌ No	✅ Yes	✅ Yes
Manage Crop Master data	❌ No	❌ No	❌ No	✅ Yes
Bulk farmer import	❌ No	❌ No	❌ No	✅ Yes
Create / manage staff accounts	❌ No	❌ No	❌ No	✅ Yes
Access audit logs	❌ No	❌ No	❌ No	✅ Yes
Configure territory hierarchy	❌ No	❌ No	❌ No	✅ Yes

7. **How should users log in?**

   Tip: eg. : Email & Password, User ID

Login method: OTP (One-Time Password) via SMS to the staff member's registered mobile number.

- There are no username/password combinations to forget or mismanage.
- On the app open, the user enters their registered mobile number.

- An OTP is sent via SMS to that number.
- On successful OTP entry, a JWT session token is issued.
- Sessions expire after 15 minutes of inactivity to protect data on shared or lost devices.
- 5 consecutive failed OTP attempts lock the account for 30 minutes; admin can unlock early.
- Roles are assigned at account creation by the administrator and cannot be self-modified.
- Device-level PIN or biometric lock is enforced before accessing the app.
- Remote session invalidation is available to admins for lost or stolen devices.

Admin web interface uses the same OTP-based authentication with an additional email-based fallback for recovery.

8. **Walk me through ONE complete task step-by-step**

Tip: Choose your most important task and describe exactly what happens.
Example: "Customer places order"

- Customer opens app and browses products
- Customer adds items to cart
- Customer enters delivery address
- Customer selects payment method
- Order is confirmed and customer gets order number
- Staff receives notification about new order

Task chosen: Field Staff member logs a farm visit, updates the crop stage, and pushes a promotional video to the farmer.

Step 1 — Open App & Authenticate

- Staff members open the FFMA app on their Android smartphone.
- Enters registered mobile number → receives OTP via SMS → enters OTP → session starts.
- Home dashboard loads: Today's Overdue List, Upcoming Stage Transitions, Quick Stats.

Step 2 — Find the Farmer

- Staff taps the search bar and types the farmer's name or village.

- The farmer card appears showing: name, village, primary crop (e.g. Grapes), current stage (e.g. Berry Set), days since last visit (e.g. 'Overdue — 16 days').
- Staff taps the card to open the full Farmer Profile.

Step 3 — Review the Profile

- The profile shows tabs: Plots & Crops | Activity Log | Recommendations | Communications | Promotions Sent.
- At the top, a banner shows: 'Suggested Promotion: Grape Berry Set Fungicide Advisory Video (Marathi)'.
- Staff reviews the Activity Log — last visit was 16 days ago, last call 8 days ago.

Step 4 — Log the Visit

- Staff taps 'Log Visit' on the farmer profile.
- System auto-captures date, time, and current GPS location.
- Staff selects visit purpose: 'Crop Advisory'.
- Staff enters notes: 'Observed early signs of downy mildew on leaves. Recommended Metalaxyl spray.'
- Staff taps the camera icon to attach 2 photos of the affected leaves (geo-stamped, time-stamped).
- Staff taps 'Save Visit' — visit is stored locally if offline, synced to server on reconnection.

Step 5 — Update the Crop Stage

- Staff taps the 'Grapes — Berry Set' crop card on the Plots tab.
- Current stage: Berry Set (auto-calculated from sowing date + Crop Master inter-stage days).
- Staff sees: 'Expected transition to Veraison in 4 days'.
- Staff confirms: 'Crop has already reached Veraison today' → taps 'Advance Stage'.
- System logs: Stage changed from Berry Set → Veraison | Date | Staff identity. This log is immutable.

Step 6 — Confirmation

- The farmer's Activity Log now shows (newest first): Visit logged · Crop stage updated ·
- The 'Days Since Last Visit' counter resets to 0.
- The overdue badge is cleared from the farmer card on the home dashboard.
- The Territory Manager's dashboard is updated in real time: 1 visit logged, 1 stage change, 1 promotion sent.

9. **What are the MUST-HAVE features for day one?**

Without these, the app is useless

Without these, the app is not useful for launch:

- OTP-based login with role-based access control (enforced at API level).
- Farmer CRUD — create, view, edit, search, and filter farmer records with all required fields.
- Farmer assignment — each farmer permanently linked to one field staff member and one territory.
- Bulk farmer import — Excel upload of ~50,000 existing farmer records with validation, error reporting, and idempotent re-import.
- Plot management — add unlimited named plots per farmer, record soil type and irrigation source.
- Crop & growth stage tracking — associate crops (from Crop Master) to plots, set sowing date, auto-calculate expected next stage dates, manually update current stage, immutable stage change log.
- Crop Master — admin-managed list of crops, varieties, growth stages, and inter-stage durations.
- Visit logging — log visits with auto-captured date, time, GPS stamp, purpose, notes, and photos.
- Call logging and click-to-call — initiate calls from within the app, auto-log with outcome notes.
- Recommendations engine — create structured product recommendations linked to crop and stage; push via WhatsApp or SMS.
- WhatsApp Business API and SMS gateway integration — send messages to individual farmers and log every send.
- Overdue visit alerts — push notifications to staff listing farmers not visited within the configured norm.
- Stage change reminders — push notifications when a farmer's crop is expected to reach its next stage.

- Smart Visit Planner — daily ranked visit schedule based on overdue status and GPS proximity.
- Field Staff Dashboard — today's overdue visits, upcoming stage transitions, activity summary, quick stats.
- Management Dashboard — territory/zone-level reports: farmers by village/crop/stage, staff activity summary, overdue visit report, staff leaderboard.
- Report export to Excel and PDF.
- Offline capability — visit logging, call logging, GPS geo-tagging, crop stage update, and photo capture all work without network; synced on reconnection.
- Territory hierarchy management — admin configures Region → Zone → Territory → Staff → Farmer.
- Full audit log — all data changes tracked with user identity, timestamp, old and new values; retained 12 months.

10. What features would be nice to have later?

These can wait for version 2

These are confirmed Phase 2 priorities. The Phase 1 data architecture (GPS plot coordinates, crop stage timelines) is specifically designed to support these without schema changes:

- NDVI & Crop Health Intelligence — satellite-derived NDVI readings fetched per geo-tagged plot; NDVI trend charts on the Plot Detail screen; alert when NDVI drops below the crop-stage benchmark.
- Soil Moisture Monitoring — satellite or sensor-based readings per plot GPS; alert when moisture is outside norm for current growth stage.
- Hyper-Local Weather Intelligence — rainfall, temperature, humidity, and wind speed fetched at village or plot level; automated bulk advisory triggers (e.g. 'Rainfall >50mm → push fungicide advisory to all Grape farmers at Flowering stage in Nashik').
- Territory Satellite Map View — interactive map of all geo-tagged plots colour-coded by NDVI health band; manager filters by crop, stage, or NDVI threshold to identify at-risk plots.
- Marathi language UI — Phase 1 ships in English and Marathi; Hindi and Kannada in Phase 2.
- Weather-event bulk push — automated promotional content triggered by configured weather event thresholds.
- Automated crop health alerts — AI-driven detection of NDVI anomalies compared to crop-stage benchmarks.

- Advanced analytics — historical trend analysis, crop yield correlation, territory growth forecasting.
- Digital Library — a centralized repository of ready-to-use content including crop schedules, marketing taglines, videos, and social media promotional materials for easy access and quick sharing by field teams.

11. **What data/information needs to be stored?**

What information do you need to track and save?
Example: Customer names, phone numbers, order details, payment status, inventory levels, etc.

The system stores data across four master data entities:

Farmer Master

- Full Name, Mobile Number (primary), Alternate Mobile, Village / Town, Taluka, District, PIN Code, State.
- Assigned Field Staff (FK), Territory (FK), Farmer Photo, Preferred Language, Land Holding (Acres).
- System-generated: Farmer ID, Date Added, Source (bulk import or in-app), Status (Active / Inactive / Transferred).

Crop & Stage Data (per Plot)

- Crop (FK to Crop Master), Variety / Hybrid Name, Planting Date, Sowing / Transplanting Date, Current Growth Stage, Previous Crop.
- System-generated: Expected Next Stage Date (auto-calculated), Stage Change Log (immutable).

Crop Master (Admin-Managed)

- Crop Name, Crop Category, Scientific Name, Varieties, Crop Schedule PDF, Reference Image, Status.
- Per Stage: Stage Name, Sequence Number, Days from Previous Stage, Stage Description.

Activity & Communication Log (per Farmer)

- Field Visits: Date, Time, GPS stamp of staff, Visit Purpose, Notes, Photos.
- Calls Logged: Date, Time, Duration, Direction (Outgoing / Incoming), Outcome Notes.
- Recommendations Given: Product Name, Dose, Timing, Method, Notes, Crop & Stage at time of recommendation.
- Promotions Pushed: Content title, Channel (WhatsApp / SMS), Timestamp, Status (Sent / Delivered / Failed), Sending Staff.
- System-calculated: Days Since Last Visit, Days Since Last Call.

Sales Person (Field Staff) Master

- Full Name, Employee ID, Mobile Number, Email, Territory (FK), Reporting Manager (FK), Last Login Timestamp, Role.

Territory Master

- Territory Name, Region / Zone Hierarchy, Assigned Field Staff, Manager Assigned, Rollup Farmer Count (auto-aggregated), Territory Map View.

Promotion Library

- Content Title, Content Type (Video / Image / PDF / Link), File / URL, Crop Tags, Stage Tags, Language Tags, Expiry Date, Status (Active / Inactive), Pre-composed WhatsApp/SMS template.

System Logs

- All data changes: Entity, Field Changed, Old Value, New Value, User Identity, Timestamp — retained 12 months.
- All bulk export events: User, Timestamp, Record Count.

12. **Does anyone need to approve certain actions?**

Specify what needs approval and who approves
Example:
- Refunds above ₹500 → Manager approval require
- New vendor registration → Owner approval required
- No approval needed for: _____

The following approval workflows are defined:

Action	Triggered By	Approved By	Notes
Bulk campaign send (zone-level)	Zonal Manager	Zonal Manager self-approves after previewing recipient list	Must preview farmer count before confirming
New staff account creation	HR / Admin request	System Administrator	Role assigned at creation; cannot be self-modified
Territory restructuring	Manager request	System Administrator	Full audit log generated on any territory change
Bulk farmer import (>1,000 records)	Admin user	Admin reviews error report before committing	Idempotent: duplicates flagged, not auto-merged
WhatsApp / SMS template registration	Admin	Meta (WABA) / TRAI (DLT)	External approval required before templates go live
Satellite API activation (Phase 2)	Admin	Management sign-off	Dependent on third-party provider pricing agreement

No approval required for:

- Individual farm visit logging by field staff.
- Individual call logging.
- Creating or editing a farmer record (within own territory).
- Sending a recommendation to an individual farmer.
- Pushing a content item to an individual farmer.

- **Bulk push to the staff member's own farmers by crop + stage (subject to WhatsApp template approval).**

13. **What should happen automatically?**

Should the system automatically send messages, reminders, or update status?

The following actions are system-triggered without any manual intervention:

Automated Action	Trigger Condition	Who Receives It
Overdue visit push notification	Days Since Last Visit exceeds configured norm (e.g. 7 or 14 days)	Assigned Field Staff
Stage transition reminder	Expected Next Stage Date reached for any farmer's crop	Assigned Field Staff
Smart Visit Planner daily refresh	Every morning at a configurable time	Field Staff sees updated ranked visit list
Farmer assignment confirmation	Bulk import completed successfully	Admin sees confirmation + any farmers visible in staff's app within 5 min
Duplicate mobile number detection	New farmer creation or import with an existing mobile number	Creating staff / admin sees a warning before saving
Stage auto-advance proposal	Expected Next Stage Date is reached	Field Staff sees a confirmation prompt; must confirm to advance
Session expiry	15 minutes of inactivity on the app	User is logged out automatically

Account lock	**5 consecutive failed OTP attempts**	**User is locked for 30 minutes; admin notified**
WhatsApp / SMS send retry	**Message fails due to temporary gateway error**	**System retries automatically; final status logged**
Offline sync	**Network connection restored after offline period**	**All queued visits, calls, photos, stage updates pushed to server within 60 sec**
[Phase 2] Weather alert bulk push	**Configured weather threshold crossed for a crop / stage / territory combination**	**Farmers in the target group receive WhatsApp/SMS advisory**
[Phase 2] NDVI drop alert	**NDVI reading falls significantly below crop-stage benchmark**	**Assigned Field Staff receives alert with plot details**

14. Where will users primarily use this app?

Eg : Mobile phone (most important)
Desktop/Laptop (most important)
Both equally
Tablet

Usage varies by role:

User Type	Primary Device	Usage Context	Key Requirement
Field Staff / Agronomist	Android smartphone (primary),	Rural farm fields — often with poor or no network	Full offline capability; data-efficient; works

			on 2GB RAM, Android 8+ devices
Territory Manager	Android smartphone (primary),	Office + occasional field visits	Dashboard and reports must be accessible on both mobile and web
Zonal / Regional Manager	Desktop/Laptop browser (primary)	Office-based reporting and planning	Admin web interface; report export to Excel and PDF
System Administrator	Desktop/Laptop browser	Office-based configuration	Full admin web interface; bulk import tools
Marketing / Content Team	Desktop/Laptop browser	Office-based content management	Content upload and tagging via admin web interface

The mobile app is the primary product for 60+ field staff and must work flawlessly in low-connectivity rural environments. The admin web interface serves all manager, admin, and content team functions and requires a stable internet connection.

15. **Do you have existing branding? Any specific design preferences?**

Colors, logo, or design style you want to follow

Company Name	**Dhanashree Crop Solutions**
Website	**www.dhanashreecropsolutions.com**

Industry	Agricultural inputs — plant nutrition and crop protection
Suggested Palette	Green as the primary brand colour (reflecting agriculture and growth); white and clean neutrals for backgrounds; high-contrast text for outdoor readability in bright sunlight.
Font	Clear, modern sans-serif (e.g. Roboto or Noto Sans) — legible on small smartphone screens in direct sunlight.
Logo	To be provided by Dhanashree team before design sprint begins. Should appear on the app splash screen, login page, and report headers.
Design Style	Clean, practical, field-worker-oriented. Minimal clutter; large tap targets (minimum 48×48dp) for use with rough hands or gloves in the field.
Language	English and Marathi (Phase 1 default) Hindi z(Phase 1.5).
Accessibility	Adjustable font size in app settings; high-contrast mode for outdoor use in bright sunlight.
App Size	Maximum 50 MB APK / IPA for initial install — important for low-storage devices common among field staff.

16. Anything else we should know?

Any other requirements, concerns, or information that would help us

Regulatory & Compliance Requirements

- WhatsApp Business API: Dhanashree must obtain a Meta-approved WhatsApp Business Account (via an approved BSP such as Interakt, AiSensy, or Gupshup) before the communication module goes live. All bulk WhatsApp sends must use pre-approved Business API templates.
- SMS DLT Compliance: All SMS bulk campaigns must use TRAI DLT-registered message templates and sender IDs. The admin is responsible for template pre-registration.
- Indian IT Act 2000: Farmer personal data (name, mobile number, location) is treated as PII — access is role-restricted and all access is logged.
- Data Residency: All data must be hosted on AWS or Azure Mumbai region (ap-south-1) for Indian data residency compliance.

Technical Constraints

- The mobile app must run on low-end Android smartphones pwith 2 GB RAM and Android 8+ (API 26+), which is common among field staff.
- Network connectivity in target rural geographies ranges from 2G to 5G. The app must be data-efficient and never require a fast connection for core workflows.
- All API endpoints must validate role permissions server-side. Data isolation cannot rely on client-side filtering alone.
- Offline sync must be idempotent — if the same visit or call is queued and synced multiple times, only one record must be created.
- The bulk import process must be idempotent: re-running the same Excel file must not create duplicate records; a 'merge' mode updates existing records matched by mobile number.

Assumptions Requiring Dhanashree Action Before Development Begins

- Dhanashree will provide the product catalogue (product names and descriptions) to be used in the Recommendations module.
- Dhanashree's agronomy team will fully define and validate the Crop Master data (crop list, stage names, inter-stage durations in days) before Phase 1 go-live.
- The ~50,000 farmer Excel sheet will be cleansed and mapped to the import template with reasonable effort. Dhanashree confirms that every farmer record includes at minimum: Full Name, Mobile Number, Village, and a valid Staff ID.
- Each field staff member has access to an Android or iOS smartphone with a working SIM card.
- Admin and manager users have access to a desktop or laptop with a browser for the web interface.

Scalability Requirements

- The system architecture must support growth to 200 field staff members and 200,000 farmers without requiring a re-architecture.
- Availability target: 99.5% uptime during business hours (6 AM – 10 PM IST); scheduled maintenance only in overnight windows.
- The system must support all 60+ field staff active and syncing simultaneously without performance degradation.

Performance Benchmarks

Metric	Target
Farmer list load (4G)	< 2 seconds for first 50 records
Photo upload (5 MB, 4G)	< 10 seconds; background upload with retry
Offline sync on reconnection	All queued actions synced within 60 seconds
Dashboard load time (4G)	< 3 seconds for current-period data
Bulk campaign (5,000 messages)	Non-blocking UI; background job with status screen
App install size	< 50 MB APK / IPA

Data Backup & Recovery

- Daily automated backups with 30-day retention.
- Point-in-time recovery capability required.
- Audit logs retained for a minimum of 12 months; accessible only to administrators.

Phase 2 Dependency

The satellite and AI capabilities (NDVI, soil moisture, weather triggers) and digital content store planned for Phase 2 are dependent on the availability and commercial pricing of third-party satellite data provider APIs (e.g. Cropin, SatSure, ISRO Bhuvan, SkyMet) for Maharashtra and the other target states. This dependency should be evaluated and a provider selected during Phase 1 development so that API contracts are in place before Phase 2 begins.